

Supervision

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| <div>Student</div> <div>Supervisor</div> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Posted at                                | 03/10/2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Main activities                          | <p>During this month, the primary focus of the internship was the analysis and documentation of the Oversight Cloud (OC) platform, including its architecture, workflows, and operational usage within the company.</p> <p>The first phase consisted of studying the existing architecture and system design of Oversight Cloud. The platform is a web-based system used to manage LiDAR-related assets, including organizations, sites, simulations, documentation, and files.</p> <p>The architecture is based on a Vue 3 frontend communicating with a Node.js/Express backend, with Airtable used as the primary database and S3-compatible storage used for file storage. Sessions are stored in Redis and the system is deployed through Docker with Traefik acting as a reverse proxy.</p> <p>A detailed technical documentation of the platform was produced, including:</p> <ul style="list-style-type: none"><li>The overall system architecture and infrastructure</li><li>The data model and entity relationships</li><li>The authentication and user management flow</li><li>The main operational workflows (account creation, simulation creation, file upload/download)</li></ul> <p>Another key activity was conducting interviews with internal users, including engineers and operational teams, to understand how Oversight Cloud is used in practice. These interviews helped identify the main use cases of the platform, which currently include:</p> <ul style="list-style-type: none"><li>Simulation storage and sharing</li><li>Access to technical documentation</li><li>Collaboration with external partners and customers</li></ul> <p>Additionally, a user experience survey was distributed to collect feedback from different internal roles, including Field Application Engineers, Customer Engineers, and Sales Engineers. The survey collected insights about user satisfaction, common workflows, and major usability issues.</p> <p>Finally, a technical analysis of the platform infrastructure and API usage was conducted. This included reviewing API endpoint traffic, identifying unused endpoints, and analyzing infrastructure costs such as DigitalOcean storage and Airtable subscription costs.</p> |
| Achievement(s)                           | <p>Several concrete deliverables were produced during this period.</p> <p>First, a complete technical documentation of the Oversight Cloud architecture was created. This documentation consolidates previously scattered information into a single knowledge base describing the system architecture, components, and technical stack.</p> <p>Second, a platform usage analysis was completed based on interviews and survey data. This helped clarify how the platform is actually used by internal teams and customers. The analysis confirmed that the core functionality of the platform is simulation management, while other features such as firmware management or file sharing are rarely used.</p> <p>Third, several technical reports were produced, including:</p> <ul style="list-style-type: none"><li>An API usage density report, identifying which API endpoints receive the most traffic and which features are actively used.</li><li>A ghost endpoint analysis, highlighting endpoints defined in the API specification but never used in production logs.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

An infrastructure cost analysis, showing that Airtable represents the largest monthly cost component, with an estimated cost of around €1000 per month.

Another important achievement was the creation of a complete Outsight Cloud presentation, summarizing the platform architecture, workflows, user feedback, and potential improvement directions.

These deliverables provide a clear overview of the current state of Outsight Cloud and establish a foundation for future architectural and product improvements.

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**Troubleshooting(s)**

During the analysis phase, several challenges were encountered while completing the assigned tasks.

One difficulty was the lack of centralized and updated documentation, which made it necessary to gather information from multiple sources such as code repositories, Confluence pages, and internal discussions to reconstruct the system architecture.

Another challenge was understanding the data model and permission system, which relies on Airtable and a permission-based structure linking users, organizations, and roles. This required additional analysis of both the database structure and backend logic before it could be properly documented.

Finally, distinguishing actual platform usage from unused features required combining user interviews, survey results, and API log analysis to identify which functionalities are actively used in practice.

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**Next step(s)**

Based on the analysis conducted during this period, the next steps of the internship will focus on proposing and evaluating improvements for Outsight Cloud.

One of the main directions is to investigate the migration of the data layer from Airtable to a relational database such as PostgreSQL, which would provide better performance, scalability, and data integrity.

Another key area of work will be the integration of a CIAM / Single Sign-On system, allowing users to authenticate using external identity providers and simplifying account management. Further work will also focus on improving the user experience of the platform, particularly:

Automating user creation and management

Improving simulation organization and versioning

Enhancing platform stability and performance

Clarifying the organization hierarchy and navigation

Finally, a long-term architectural analysis will be conducted to evaluate a potential migration of the infrastructure to a cloud-native architecture, possibly using AWS services to improve scalability and maintainability.

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**Intermediate report  
(only one per month)**

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